

Practice Quiz 09C — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 09

Well-definedness on a quotient; onto maps and inverses.

Write $[x]_8$ for the congruence class of x modulo 8.

1. Consider

$$f : \mathbf{Z}_8 \longrightarrow \mathbf{Z}_8, \quad f([x]_8) = [3x + 1]_8.$$

(a) Prove that f is well defined.

Solution Suppose $[x]_8 = [y]_8$. Then $8 \mid (x - y)$, so $x - y = 8k$ for some integer k . Therefore

$$(3x + 1) - (3y + 1) = 3(x - y) = 8(3k).$$

Thus $8 \mid ((3x + 1) - (3y + 1))$, so $[3x + 1]_8 = [3y + 1]_8$. Hence the output does not depend on the representative and f is well defined. ■

(b) Prove that f is onto and determine f^{-1} .

Solution The number 3 is an inverse of 3 modulo 8, since $3 \cdot 3 \equiv 1 \pmod{8}$. Given an arbitrary $[y]_8$, choose

$$[x]_8 = [3(y - 1)]_8.$$

Then

$$f([x]_8) = [3 \cdot 3(y - 1) + 1]_8 = [y]_8,$$

because $3 \cdot 3 \equiv 1 \pmod{8}$. Thus f is onto. The calculation also gives

$$f^{-1}([y]_8) = [3(y - 1)]_8.$$

The same representative check as in part (a), with coefficient 3, shows that this inverse rule is well defined. ■